

Big Data

Concepts, Technologies & Ecosystem — Complete Notes

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1. Introduction to Big Data

Big Data refers to extremely large and complex datasets that traditional data processing software cannot manage, capture, or analyze efficiently within a reasonable time frame. It comes from sources like social media, sensors, transactions, mobile devices, and machine logs.

Why Big Data Matters

- Enables data-driven decision making for businesses
- Helps discover hidden patterns, trends, and correlations
- Powers recommendation systems, fraud detection, and predictive maintenance
- Drives advances in AI and machine learning through large training datasets

2. Characteristics of Big Data (5 V's)

V	Meaning	Example
Volume	Massive amount of data generated	Petabytes of social media data daily
Velocity	Speed at which data is generated/processed	Real-time stock market feeds
Variety	Different formats/types of data	Text, images, video, sensor data
Veracity	Quality/trustworthiness of data	Filtering noisy or inconsistent data
Value	Usefulness extracted from data	Business insights, better decisions

3. Types of Big Data

Structured Data

Data organized in a fixed format such as rows and columns — e.g. relational database tables, spreadsheets.

Semi-Structured Data

Data that doesn't fit a rigid schema but has some organizational structure — e.g. JSON, XML, log files.

Unstructured Data

Data with no predefined structure — e.g. images, videos, audio, social media posts, emails. This makes up the majority of big data.

4. Big Data Architecture

A typical big data pipeline consists of several layers that move data from raw collection to actionable insight.

Data Sources

- > Ingestion Layer (Kafka, Flume, Sqoop)
- > Storage Layer (HDFS, S3, NoSQL DBs)
- > Processing Layer (MapReduce, Spark, Flink)
- > Analytics Layer (Hive, Pig, ML models)
- > Visualization (Tableau, Power BI, dashboards)

Batch vs Stream Processing

- Batch processing — handles large volumes of data at scheduled intervals (e.g. Hadoop MapReduce)
- Stream processing — handles data in real-time as it arrives (e.g. Apache Kafka, Spark Streaming, Flink)

5. Hadoop Ecosystem

Apache Hadoop is an open-source framework for distributed storage and processing of large datasets across clusters of commodity hardware.

Component	Purpose
HDFS	Distributed storage system
MapReduce	Distributed data processing model
YARN	Resource management & job scheduling
Hive	SQL-like querying on top of Hadoop
Pig	Scripting platform for data analysis
HBase	NoSQL database on top of HDFS
Sqoop	Transfers data between Hadoop and RDBMS
Flume	Collects and moves streaming log data
Zookeeper	Coordination service for distributed apps
Oozie	Workflow scheduler for Hadoop jobs

6. HDFS (Hadoop Distributed File System)

HDFS stores huge files by splitting them into fixed-size blocks (default 128MB) distributed and replicated across multiple machines in a cluster for fault tolerance.

Key Components

- NameNode — master node that stores metadata (file names, block locations)
- DataNode — worker nodes that store the actual data blocks
- Block replication — default replication factor of 3 for fault tolerance
- Secondary NameNode — periodically checkpoints the NameNode's metadata

```
# Common HDFS commands
hdfs dfs -ls /user/data          # list files
hdfs dfs -put localfile.txt /data # upload file
hdfs dfs -get /data/file.txt .    # download file
hdfs dfs -rm /data/file.txt       # delete file
```

7. MapReduce

MapReduce is a programming model for processing large datasets in parallel across a distributed cluster, consisting of two main phases.

Map Phase

Input data is split into chunks and processed in parallel, producing key-value pairs.

Reduce Phase

Key-value pairs with the same key are grouped and aggregated to produce the final output.

```
# Word Count Example (pseudocode)
map(document):
    for word in document:
        emit(word, 1)

reduce(word, counts):
    emit(word, sum(counts))
```

8. Apache Spark

Apache Spark is a fast, in-memory distributed data processing engine, often used as a faster alternative to MapReduce (can be up to 100x faster for certain workloads).

Key Features

- In-memory computation — reduces disk I/O compared to MapReduce
- Supports batch, streaming, machine learning (MLlib), and graph processing (GraphX)
- APIs available in Python (PySpark), Scala, Java, and R
- Core abstraction: RDD (Resilient Distributed Dataset) and DataFrames

```
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("Demo").getOrCreate()
df = spark.read.csv("data.csv", header=True)
df.groupBy("category").count().show()
```

9. NoSQL Databases

NoSQL databases are designed to handle unstructured/semi-structured data at scale, offering flexible schemas unlike traditional relational databases.

Type	Example DBs	Use Case
Key-Value	Redis, DynamoDB	Caching, session storage
Document	MongoDB, CouchDB	Content management, catalogs

Column-family	Cassandra, HBase	Time-series, large-scale writes
Graph	Neo4j, ArangoDB	Social networks, recommendation engines

10. Data Warehousing vs Data Lake

Aspect	Data Warehouse	Data Lake
Data Type	Structured only	Structured, semi & unstructured
Schema	Schema-on-write	Schema-on-read
Purpose	Business intelligence, reporting	Big data, ML, exploratory analysis
Cost	Higher (processed storage)	Lower (raw storage)
Examples	Amazon Redshift, Snowflake	Amazon S3, Hadoop HDFS

11. Big Data Analytics Lifecycle

- Data Discovery — understand business problem and available data sources
- Data Preparation — collect, clean, and transform raw data (ETL)
- Model Planning — choose appropriate analytical techniques
- Model Building — develop and train models on training datasets
- Communicate Results — present findings via reports and visualizations
- Operationalize — deploy models into production systems

12. Tools & Technologies Overview

Category	Popular Tools
Storage	HDFS, Amazon S3, Google Cloud Storage
Processing	Hadoop MapReduce, Apache Spark, Apache Flink
Streaming	Apache Kafka, Apache Storm, Spark Streaming
Querying	Hive, Presto, Apache Drill
Visualization	Tableau, Power BI, Apache Superset
Workflow	Apache Airflow, Oozie
Cloud Platforms	AWS EMR, Google BigQuery, Azure HDInsight

13. Applications & Challenges

Real-World Applications

- Healthcare — predictive diagnostics, patient monitoring

- Retail & E-commerce — personalized recommendations, demand forecasting
- Finance — fraud detection, algorithmic trading, risk analysis
- Transportation — route optimization, predictive maintenance
- Social Media — sentiment analysis, targeted advertising

Key Challenges

- Data storage and scalability as volumes keep growing
- Ensuring data quality, privacy, and security
- Integrating data from multiple heterogeneous sources
- Talent shortage for skilled big data engineers/scientists
- High infrastructure and processing costs

Big Data governance and privacy regulations (like GDPR) are becoming increasingly important as organizations collect more personal data.

End of Notes — Big Data is a vast and fast-evolving field. Use these notes as a foundation, and explore hands-on practice with Hadoop/Spark for deeper understanding.